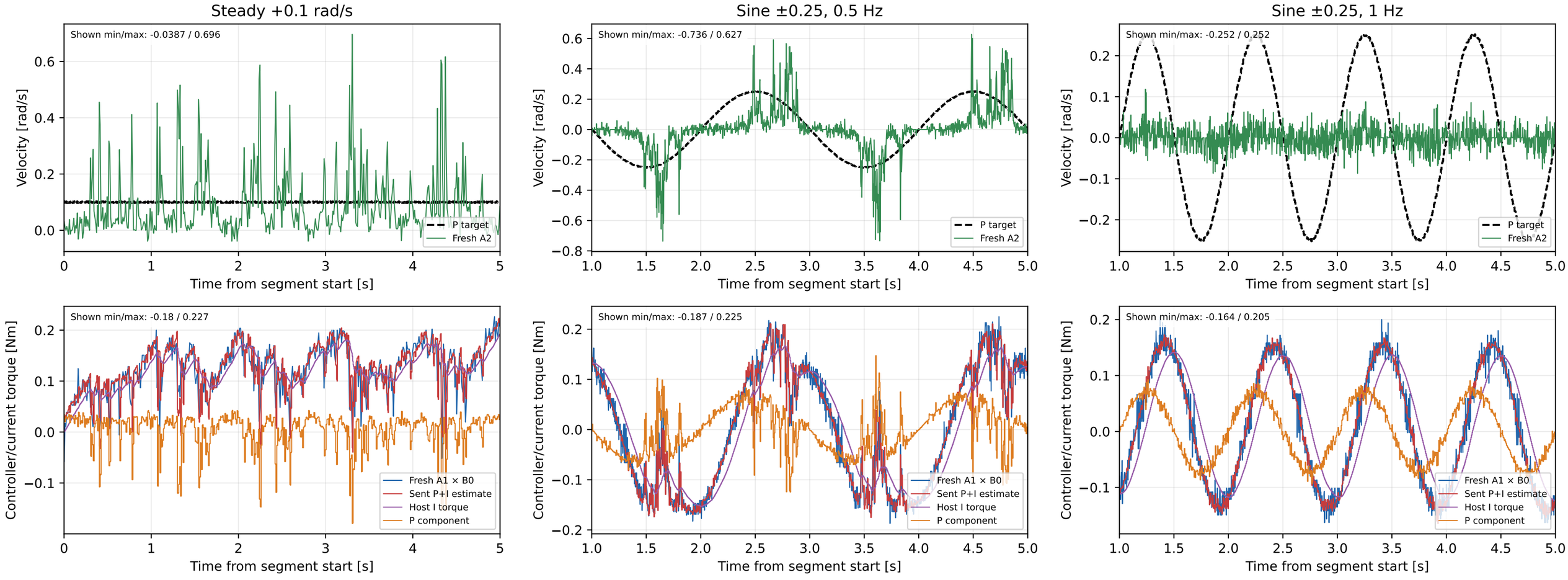

MIT PI torque decomposition: integral supports movement but cannot cure every reversal
Steady +0.1 rad/s
Sine ±0.25, 0.5 Hz
Sine ±0.25, 1 Hz
Shown min/max: -0.0387 / 0.696
Shown min/max: -0.736 / 0.627
Shown min/max: -0.252 / 0.252
Velocity [rad/s]
Time from segment start [s]
P target
Fresh A2
Shown min/max: -0.18 / 0.227
Shown min/max: -0.187 / 0.225
Shown min/max: -0.164 / 0.205
Controller/current torque [Nm]
Fresh A1 × B0
Sent P+I estimate
Host I torque
P component
Command estimates use latest A2 at send; A1 is independently queried, unsmoothed. Known-law held-out current R²=0.917; this is not forward plant prediction.